

Mixed-Sign Stochastic Neural Fields: Full Exponential Ergodicity Without Kernel Definiteness

Automated mathematics run `fa_banach_001`, lane 1

26 August 2026

Status: full resolution of the stated mixed-kernel exponential- ergodicity problem, likely valid. In the source’s quantitative ergodicity regime, its invariant-measure, exponential-ergodicity, and exponential-mixing conclusions hold for arbitrary mixed-sign kernels. The sign and compact-subspace assumptions are unnecessary.

Abstract

Otsetova and Tölle identify the extension of their main exponential- ergodicity result to kernels mixing excitation and inhibition as an open problem. We resolve that question in the same quantitative regime. If

$$\kappa := 2\alpha - 2\|K\| \operatorname{Lip}(f) - \operatorname{Lip}(B)^2 > 0,$$

then the Markov semigroup contracts the quadratic Wasserstein metric by $e^{-\kappa t/2}$, has a unique invariant law with finite second moment, and converges exponentially to it from every finite-second-moment initial law. The proof uses synchronous coupling and a uniform Itô moment estimate, so it does not use any sign or symmetry of the kernel, nor the source’s compact nonlocal Hilbert subspace. A direct corollary recovers the source’s main conclusions under the dimensionally consistent version of its Assumption 5, with Assumptions 3 and 4 deleted.

1 Source question

The source is A.-M. Otsetova and J. M. Tölle, *Ergodicity for stochastic neural field equations*, arXiv:2505.14012v1 (2025)[1]. On PDF page 5, after describing their sign-definite theory, they write:

“It would be interesting to obtain similar results, when the kernel is neither non-negative or non-positive definite, relating to mixed inhibition and excitation.”

Immediately before the open sentence, the source describes its main contribution as existence of a unique invariant measure and exponential ergodicity under a quantitative relation between decay, the kernel, the activation, and the noise. Thus the request concerns the exponential- ergodicity regime of Theorem 1.2(iii); the following sentence identifies metastability and Kramers’ law as a separate open problem.

The source’s existence argument restricts the dynamics to a compactly embedded nonlocal Hilbert subspace built from a positive or negative part of the kernel. That route is unavailable for a general mixed-sign operator. The result below replaces compactness by strict metric contraction.

Assumption 4 in Subsection 4.1. The condition resembles the assumptions for the exponential frame in [2, 15, 18] and displays the natural hierarchy of parameters of the model, see also [33].

It would be interesting to obtain similar results, when the kernel is neither non-negative or non-positive definite, relating to mixed inhibition and excitation. Another interesting open problem is the analysis of metastability and Kramers' law [3] which could be based on the gradient structure introduced in [32].

Figure 1: The open direction in the source paper, PDF page 5.

2 Setting and statement

Let $\mathbf{H} = L^2(U, \rho)$, and consider the source equation

$$dX_t = (-\alpha X_t + KF(X_t)) dt + B(X_t) dW_t, \quad X_0 = x \in \mathbf{H}, \quad (1)$$

where $F(u)(\xi) = f(u(\xi))$. We retain Assumptions 1 and 2 of the source. In particular, $\mathbf{H}$ is separable, $K \in \mathcal{L}(\mathbf{H})$, $f : \mathbb{R} \rightarrow \mathbb{R}$ is Lipschitz, and $B : \mathbf{H} \rightarrow L_2(V, \mathbf{H})$ is Lipschitz. The source's Theorem 3.3 then gives a unique global strong solution and its time-homogeneous Markov semigroup.

Set

$$L_f := \text{Lip}(f), \quad L_B := \text{Lip}(B), \quad \kappa := 2\alpha - 2\|K\|L_f - L_B^2.$$

No self-adjointness or quadratic-form sign is imposed on K .

Theorem 1 (Mixed-kernel contraction theorem). *Assume $\kappa > 0$. Then:*

1. *for all $\nu, \eta \in \mathcal{P}_2(\mathbf{H})$ and $t \geq 0$,*

$$W_2(P_t^* \nu, P_t^* \eta) \leq e^{-\kappa t/2} W_2(\nu, \eta); \quad (2)$$

2. *there is a unique invariant measure $\mu \in \mathcal{P}_2(\mathbf{H})$;*

3. *every $\nu \in \mathcal{P}_2(\mathbf{H})$ satisfies*

$$W_2(P_t^* \nu, \mu) \leq e^{-\kappa t/2} W_2(\nu, \mu); \quad (3)$$

in particular, for every Lipschitz $\phi : \mathbf{H} \rightarrow \mathbb{R}$ for which the integrals exist,

$$\left| P_t \phi(x) - \int_{\mathbf{H}} \phi d\mu \right| \leq \text{Lip}(\phi) e^{-\kappa t/2} \left(\int_{\mathbf{H}} \|x - y\|^2 \mu(dy) \right)^{1/2}.$$

The conclusions hold for kernels with arbitrary mixtures of positive and negative spectrum.

Corollary 2 (Exact mixed-kernel extension of the source theorem). *Retain source Assumptions 1 and 2, allow K to have arbitrary mixed positive and negative spectrum, and write L_B for the Lipschitz constant specified in source Assumption 2. If the dimensionally consistent form of the source's quantitative Assumption 5 holds,*

$$2\sqrt{2} \|K\|L_f + L_B^2 < 2\alpha, \quad (4)$$

then all conclusions of Theorem 1 hold. In particular, the unique invariant law with second moment exists and the semigroup is exponentially ergodic and exponentially mixing, without source Assumptions 3 and 4.

Proof. Condition (4) implies

$$2\alpha - 2\|K\|L_f - L_B^2 > 2(\sqrt{2} - 1)\|K\|L_f \geq 0.$$

Hence $\kappa > 0$, and Theorem 1 applies. The theorem's W_2 estimate implies both weak mixing and the bounded-Lipschitz (Fortet–Mourier) estimate used in the source. $\square$

3 Proof intuition

The sign assumption in the source is used to manufacture compactness, not to control the distance between two solutions. Drive two solutions by the same Wiener process. The linear decay contributes -2α to the evolution of their squared distance, while the neural interaction costs at most $2\|K\|L_f$ and the quadratic variation costs L_B^2 . Thus $\kappa > 0$ gives a strict mean-square contraction.

Contraction alone gives uniqueness but not existence. A second Itô estimate absorbs the affine terms $f(0)$ and $B(0)$, giving a second-moment bound uniform in time. The semigroup identity then makes $P_t^*\delta_0$ a Wasserstein-Cauchy family: evolve its earlier and later laws for their common final time and contract the initial discrepancy. Completeness of $(\mathcal{P}_2(\mathbf{H}), W_2)$ supplies the invariant limit.

4 Proof

4.1 Synchronous contraction

Let X_t^x and X_t^y solve (1) with initial conditions x, y , driven by the same cylindrical Wiener process, and write $D_t = X_t^x - X_t^y$. Applying Itô's formula to $\|D_t\|^2$, localizing the martingale if necessary, and taking expectations gives

$$\begin{aligned} \frac{d}{dt}\mathbb{E}\|D_t\|^2 &= -2\alpha\mathbb{E}\|D_t\|^2 + 2\mathbb{E}\langle D_t, K(F(X_t^x) - F(X_t^y)) \rangle \\ &\quad + \mathbb{E}\|B(X_t^x) - B(X_t^y)\|_{L_2(V, \mathbf{H})}^2. \end{aligned}$$

The Nemytskii map has Lipschitz constant L_f . Therefore

$$2\langle D_t, K(F(X_t^x) - F(X_t^y)) \rangle \leq 2\|K\|L_f\|D_t\|^2,$$

and the noise term is bounded by $L_B^2\|D_t\|^2$. Hence

$$\frac{d}{dt}\mathbb{E}\|D_t\|^2 \leq -\kappa\mathbb{E}\|D_t\|^2,$$

so Grönwall's lemma yields

$$\mathbb{E}\|X_t^x - X_t^y\|^2 \leq e^{-\kappa t}\|x - y\|^2. \tag{5}$$

For any coupling of initial laws ν, η , use the coupled initial conditions and the same Wiener process. Taking the infimum over initial couplings in (5) proves (2).

4.2 A uniform second moment

All three alternatives in source Assumption 1 have $m_\rho := \int_U \rho < \infty$. Put

$$c_f := |f(0)|\sqrt{m_\rho}, \quad b_0 := \|B(0)\|_{L_2(V, \mathbf{H})}.$$

Then

$$\|F(u)\| \leq L_f \|u\| + c_f, \quad \|B(u)\|_{L_2} \leq L_B \|u\| + b_0.$$

Choose $\varepsilon > 0$ so that

$$q := 2\alpha - 2\|K\|L_f - (1 + \varepsilon)L_B^2 > 0;$$

such a choice exists because $\kappa > 0$. Itô's formula and $(a + b)^2 \leq (1 + \varepsilon)a^2 + (1 + \varepsilon^{-1})b^2$ give, for $M_x(t) := \mathbb{E}\|X_t^x\|^2$,

$$\begin{aligned} M'_x(t) &\leq -qM_x(t) + 2\|K\|c_f\sqrt{M_x(t)} + (1 + \varepsilon^{-1})b_0^2 \\ &\leq -\frac{q}{2}M_x(t) + \frac{2\|K\|^2c_f^2}{q} + (1 + \varepsilon^{-1})b_0^2. \end{aligned}$$

Define

$$\gamma := \frac{q}{2}, \quad C_0 := \frac{2\|K\|^2c_f^2}{q} + (1 + \varepsilon^{-1})b_0^2.$$

Another application of Grönwall gives

$$M_x(t) \leq e^{-\gamma t} \|x\|^2 + \frac{C_0}{\gamma} \quad (t \geq 0). \quad (6)$$

In particular, the laws $\nu_t := P_t^* \delta_0$ have uniformly bounded second moments.

4.3 Existence, invariance, and uniqueness

Let $r \geq t$ and write $r = t + s$. By the semigroup property and (2),

$$\begin{aligned} W_2(\nu_r, \nu_t) &= W_2(P_t^* \nu_s, P_t^* \delta_0) \\ &\leq e^{-\kappa t/2} W_2(\nu_s, \delta_0) \\ &\leq e^{-\kappa t/2} \left(\sup_{a \geq 0} \int_{\mathbf{H}} \|z\|^2 \nu_a(dz) \right)^{1/2}. \end{aligned}$$

The last supremum is finite by (6). Thus $(\nu_t)_{t \geq 0}$ is Cauchy as $t \rightarrow \infty$ in W_2 . Since $\mathbf{H}$ is complete and separable, $(\mathcal{P}_2(\mathbf{H}), W_2)$ is complete[2]; hence $\nu_t \rightarrow \mu \in \mathcal{P}_2(\mathbf{H})$.

For fixed $h \geq 0$, contraction makes P_h^* continuous on $\mathcal{P}_2(\mathbf{H})$, and

$$P_h^* \mu = \lim_{t \rightarrow \infty} P_h^* \nu_t = \lim_{t \rightarrow \infty} \nu_{t+h} = \mu.$$

Thus μ is invariant. If $\tilde{\mu} \in \mathcal{P}_2(\mathbf{H})$ is another invariant law, then for every $t > 0$,

$$W_2(\mu, \tilde{\mu}) = W_2(P_t^* \mu, P_t^* \tilde{\mu}) \leq e^{-\kappa t/2} W_2(\mu, \tilde{\mu}).$$

It follows that $\mu = \tilde{\mu}$. Applying (2) with $\eta = \mu$ proves (3). Finally, Kantorovich–Rubinstein duality and $W_1 \leq W_2$ give the stated Lipschitz-observable estimate. This completes the proof.

5 Sharpness of the contraction coefficient

The squared noise constant and the half-rate are not artifacts of the proof. Let $\mathbf{H} = \mathbb{R}^2$, let

$$K = \begin{pmatrix} a & 0 \\ 0 & -b \end{pmatrix}, \quad a \geq b > 0, \quad f(r) = r,$$

and let $B(u)q = \sigma q u$ for $q \in \mathbb{R}$. The kernel has genuinely mixed spectrum, $\|K\| = a$, and $L_B = |\sigma|$. If two solutions start with a difference in the first eigendirection and are driven synchronously, their difference D_t obeys

$$dD_t = (-\alpha + a)D_t dt + \sigma D_t dW_t.$$

Consequently

$$\mathbb{E}\|D_t\|^2 = e^{-(2\alpha - 2a - \sigma^2)t} \|D_0\|^2.$$

Thus $\kappa = 2\alpha - 2\|K\|L_f - L_B^2$ is the exact mean-square exponent already for a mixed-spectrum linear example. For two same-sign initial coordinates on this eigendirection, both laws are positive scalar multiples of the same lognormal random variable; their monotone coupling is optimal in one dimension. Hence taking a square root gives the exact W_2 scale $e^{-\kappa t/2}$.

6 Verification and source comparison

The proof uses only source Assumptions 1 and 2, the operator norm of K , and standard Hilbert-space Itô calculus. In particular, it never uses the quadratic form $\langle Ku, u \rangle$, so mixed positive and negative spectrum causes no change.

Two constant conventions deserve attention. The source defines its noise constant by $\|B(u) - B(v)\|_{L_2} \leq C_B \|u - v\|$, but a quadratic-variation estimate then naturally contains C_B^2 . Also, mean-square contraction at rate $e^{-\kappa t}$ yields Wasserstein and Lipschitz-observable contraction at rate $e^{-\kappa t/2}$, after taking a square root. The statement here uses these squared/square-root conventions consistently. The separate verifier report records an adversarial step check.

7 Resolution scope and novelty status

This is a full answer to the source’s stated mixed-kernel exponential-ergodicity question in the quantitative regime in which the source formulates its main ergodicity theorem. Corollary 2 recovers that theorem’s conclusions under its corrected smallness relation while deleting precisely the two sign-dependent assumptions. The result is stronger in two respects: it assumes the sharper $\kappa > 0$ condition and gives W_2 contraction.

No claim is made about invariant measures when $\kappa \leq 0$, invariant laws outside $\mathcal{P}_2(\mathbf{H})$, or the separately stated metastability and Kramers-law problem. These are broader parameter or moment questions, not qualifications to the resolution of the source’s strict-dissipation ergodicity question.

The run’s four lightweight indexes and local parsed arXiv corpus were searched using arXiv:2505.14012, the exact phrase about mixed inhibition and excitation, the paper title, and close keywords concerning neural fields and Wasserstein contraction. Bounded web and arXiv searches used the same phrase and close variants. No separate paper explicitly resolving this source question was found, including a check of the latest official arXiv source available on 26 August 2026. The abstract synchronous-contraction mechanism is standard in dissipative stochastic evolution equations, so the novelty claim is deliberately limited to the exact sign-free resolution and its proof for this source model.

References

- [1] A.-M. Otsetova and J. M. Tölle, *Ergodicity for stochastic neural field equations*, arXiv:2505.14012v1, 2025.
- [2] C. Villani, *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften 338, Springer, 2009.